

“How Frightened Should We Be of AI”

1. Article Synopsis

In the article “How Frightened Should We Be of A.I.?”, published in *The New Yorker* in 2018, Tad Friend explores the rapid development of artificial intelligence and the potential risks that come with creating increasingly intelligent systems. The article discusses both the benefits of AI and the uncertainty surrounding its future, particularly the possibility of developing artificial general intelligence (AGI), which could eventually surpass human intelligence. While AI has the potential to improve many aspects of society, Friend raises an important question about whether humans will be able to control the technology they create.

One of the main ideas presented in the article is that AI systems do not always behave the way their developers intend. Friend provides examples of AI programs producing unexpected results, including Microsoft's Tay chatbot, which began generating offensive content after interacting with users online. He also discusses experiments where AI programs found unusual ways to complete assigned tasks, demonstrating that a system can technically accomplish its objective without behaving as its creators expected. These examples highlight the difficulty of predicting how intelligent systems will respond to different environments and situations.

Another important topic is the increasing ability of AI to generate realistic content. Friend discusses technologies that can manipulate photographs, replace faces in videos, and generate synthetic audio. Although these technologies demonstrate impressive advancements in artificial intelligence, they also create opportunities for misinformation and deception. As AI-generated media becomes more convincing, people may have a harder time determining whether the information they encounter online is authentic. This creates concerns about how technology could influence public trust and people's understanding of reality.

The article also examines the ethical responsibilities associated with developing AI. Friend discusses how AI systems can be difficult to understand because their decision-making processes are not always transparent. This becomes particularly concerning when AI is used to make decisions that affect people's lives. He also explores the possibility of AI systems pursuing goals that do not align with human intentions, even when those systems were originally designed to be helpful.

Overall, the article emphasizes that the risks of artificial intelligence do not necessarily come from technology becoming intentionally harmful. Instead, problems can

arise when developers fail to anticipate unintended consequences, misunderstand how their systems operate, or prioritize technological advancement over safety. The article encourages readers to consider not only what AI is capable of accomplishing, but also the responsibility humans have when designing and implementing these systems.

2. Application to Software Engineering

After reading this article, we identified several important lessons that apply to us as software engineers, especially regarding the responsibilities developers have when creating new technologies. One of the biggest takeaways from the article is that technology does not necessarily have to be intentionally harmful to create serious problems. Even when software is developed with good intentions, unexpected behaviors, design flaws, and a lack of proper testing can lead to unintended consequences. As future software engineers, we believe it is important to recognize these risks and consider how our decisions throughout the development process can affect the people who eventually use our software.

One of the most important connections we noticed was the possibility of software behaving differently than developers intended. Friend discusses examples of AI systems that technically accomplish their assigned tasks but produce unexpected or undesirable results. This raises an important question about how software engineers can ensure that their programs behave appropriately in situations they may not have anticipated. Even if software is functioning according to its programmed objectives, the results could still negatively affect users. In more serious situations, poorly designed objectives, insufficient safeguards, or a lack of human oversight could allow AI systems to make harmful decisions that developers struggle to control and understand behind the scenes. For example, if an AI system used in healthcare makes an incorrect diagnosis or treatment recommendation, and the software is designed to act on that decision without proper human oversight, it could put patients' lives at risk. Additionally, biased training data or algorithms could lead to discriminatory outcomes, disproportionately affecting marginalized communities, especially when AI is used to make important decisions in legal, political, and governmental contexts. To help prevent these situations, developers should carefully consider the objectives they give their systems and evaluate not only whether the software works, but also whether its behavior aligns with the intended purpose.

Another important issue is the failure of interconnected systems. Modern software applications rely on multiple components, including databases, external APIs, machine

learning models, and user interfaces. Even if each individual component works correctly, problems can arise when these systems interact. For example, an AI model might generate an inaccurate prediction, which another component then treats as reliable information and uses to make an important decision. If the system does not account for uncertainty or verify the information, one small error could spread throughout the application and lead to much larger consequences. This shows the importance of testing not only individual components but also the entire system to see how everything functions together. Implementing safeguards, validating outputs, and designing systems that can handle unexpected failures are important ways to reduce these risks.

The article also highlights the importance of thorough testing and understanding the limitations of software. As AI systems become increasingly complex, it may become more difficult for developers to predict how they will respond to every possible situation. In a software engineering environment, this means that simply testing whether a program works under normal conditions is not enough. Engineers should also consider unusual inputs, unexpected user behavior, and situations where the software might produce incorrect results. For AI applications in particular, testing should include examining biases, inaccurate predictions, and the possibility of the system producing misleading information. Regular testing, monitoring, and updating software after deployment can help developers identify problems before they cause widespread harm.

Beyond technical failures, we believe the article raises important ethical questions. In the workforce, developers may face pressure to release products quickly, meet deadlines, or prioritize new features over safety and reliability. However, rushing development or overlooking known issues could have serious consequences, especially when software is used in critical fields such as healthcare, law, and defense. Engineers have a responsibility to communicate potential risks, be transparent about the limitations of their technology, and speak up when they believe a system may be unsafe. Even if a particular decision comes from higher-level management, developers should still consider the ethical implications of the work they contribute to, as well as the potential impact of the products their company develops.

Ultimately, our biggest takeaway from this article is that successful software engineering involves much more than creating technology that functions correctly. It also requires understanding how that technology behaves with real users and recognizing as well as minimizing the potential consequences of failure. After reading this article, we believe that while developing software, it's important to prioritize responsible design, test thoroughly,

communicate properly, and maintain transparency throughout the development process.

While it may not be possible to anticipate every problem, taking these precautions can help reduce the risk of unintended harm. The article reminds us that as technology becomes more powerful, the responsibility of the people developing it becomes increasingly important.

Reference

Friend, T. (2018, May 14). *How frightened should we be of A.I.?* The New Yorker.

<https://www.newyorker.com/magazine/2018/05/14/how-frightened-should-we-be-of-ai>